METRIC
MIL-P-70956A (AR)
AMENDMENT 1
08 JANUARY 1992

MILITARY SPECIFICATION

PROPELLANT, M14, FOR USE IN CARTRIDGE 120 mm, TPCSDS-T, M865

This Amendment forms a part of Military Specification MIL-P-70956A (AR), dated 05 August 1991, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies for the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
*3	3
*4	4
4a	None.

* Indicates specific pages revised by this Amendment.

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3.1.1.1 Nitrogen content. - The nitrogen content of the nitrocellulose shall be 13.15 ± 0.05 percent as determined in accordance with 4.5.1.2.

3.1.1.2 Moisture content. - The moisture content shall be 0.60 ± 0.20 percent.

3.1.1.3 Residual solvent. - The residual solvent shall be a maximum of 0.70 percent.

3.1.1.4 Density. - Density shall be recorded for information.

3.1.1.5 Heat of explosion (HOE). - The HOE shall be recorded for information.

3.1.1.6 Relative quickness (R.Q.) and relative force (R.F.). - This test shall be conducted for information only.

3.1.1.7 Hygroscopicity. - The hygroscopicity shall not exceed 1.0 percent at 30 degrees Celsius (C) and 90 percent relative humidity. Data shall be taken for information on the first 5 lots produced.

3.1.1.8 Propellant aging. - The minimum time from the propellant packing date to the proving ground testing shall be 60 days.

* 3.1.1.9 Graphite. - The propellant shall be graphited (glazed) to 0.20 percent maximum as determined by an added basis using Grade III or IV as specified in MIL-G-155.

3.1.2 Physical requirements.

3.1.2.1 Physical properties. - The propellant shall conform to the requirements listed in Table I. Mean variations and standard deviations of the dimensions are expressed as a percentage of the mean dimension.

TABLE I. PHYSICAL PROPERTIES

*	Length (L)	1.18 - 0.13 centimeters (cm)
	Grain diameter (D)	0.58 - 0.05 cm
	Perforation diameter (d)	0.05 - 0.01 cm
	Average web (W)	0.10 + 0.01 cm
	L/D	1.90 to 2.10
	D/d	10.3 to 14.3
	Uniformity (L)	6.25% mean variation
	Uniformity (D)	4.75% mean variation

3.1.2.2 Bulk density. - The bulk density shall be recorded for information.

3.1.2.3 Clusters. - A cluster is three or more grains stuck together

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The propellant shall not contain clusters.

3.1.2.4 Physical appearance score. - The physical appearance score shall be recorded for information.

3.1.2.5 Appearance and condition. - Finished propellant shall be greenish-gray in color with smooth clean surface, clean cut right angle ends, free of sharp points, dust and foreign material.

3.1.2.6 Weight uniformity. - One hundred propellant grains shall be weighed for information as determined in accordance with 4.5.1.12.

* 3.1.2.7 Loadability. - The charge weight required to meet the ballistic performance requirements shall be 7.70 kilograms maximum.

3.2 Ballistic assessment requirements. - The propellant when fired in the M256 gun with the M865 projectile shall comply with the following requirements: Performance temperatures shall be from -46 to 63 degrees C.

3.2.1 Velocity. - The propellant shall be assessed to a velocity of 1700 meters per second (mps) at 21 degrees C.

3.2.2 Uniformity of velocity. - The standard deviation of velocity at 21 degrees C shall not exceed 6 mps.

3.2.3 Individual maximum pressure (IMP). - The propellant shall not produce an IMP greater than 6220 bars, nor shall the average pressure plus three standard deviations exceed 6220 bars at any temperature.

3.2.4 Negative differential pressure. - The absolute value of the negative differential pressure shall not exceed 345 bars.

3.2.5 Burning residue, smoldering residue, or flareback.

3.2.5.1 Burning or smoldering residue. - The propulsion ignition system shall function such that no burning or smoldering residue shall occur after breech opening in the gun chamber or breech area as observed on an infrared enhanced video recording system when tested in accordance with paragraph 4.5.2.2.2.

3.2.5.2 Flareback. - Flareback of any duration after case base extraction shall be cause for referral (see 4.5.2.2.2). Flareback is the ignition and combustion of unconsumed residual gases with flames extending aft of the breech ring.

3.4 Preproduction inspection. - This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples shall be as specified by the Contracting Officer.

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3.5 Workmanship. - The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

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